Tipuri de integrare

(1) Integrare ODE (ec. diferent. ordinare)

$[x_0, x_0+a]$

Problema Cauchy (PC) $\begin{cases} y' = f(x,y) \\ y(x_0) = \lambda \end{cases} \longrightarrow y = y(x)$, $y = [x_0, x_0+a]$

ex $f = x+y$
$f = x^2+y^2$
$f = -y$

$\forall\ x \in [x_0, x_0+a] \quad \exists\ u, v \in \mathbb{R}$ și $L \in \mathbb{R}_+^*$

a.î. $|f(x,u) - f(x,v)| < L \cdot |u-v|$

Th. constructivistă Picard

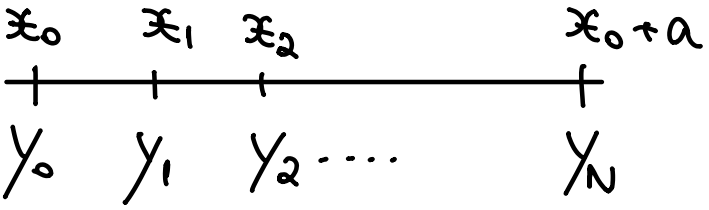
$$y(x) = \lambda + \int_{x_0}^{x} f(t, y(t))\,dt$$

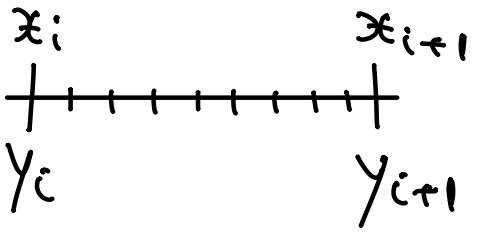
(2) Metoda Euler (P.E)

$y(x_i) \to y_i$ (y P.C., y_i P.E.) cu $x_i = x_0 + ih$, $h = \frac{a}{N}$, $i = 0:N$

$$(P.E.) \begin{cases} Y_0 = \lambda \\ Y_{i+1} = Y_i + h \cdot f(x_i, Y_i) \end{cases} \quad i = 0:N-1$$

(3) Metoda Runge-Kuta de ordin q





$x_{ij} = x_i + u_j h \quad j = 0 : q-1$

$h = x_{i+1} - x_i \quad 0 \le u_j \le 1$

q = ord. metod.

$$(RK_q) \begin{cases} y_{i0} = y_i \\ y_{ij} = y_i + h \cdot \sum_{\ell=0}^{j-1} k_{j\ell} f(x_{i\ell}, y_{i\ell}) \end{cases}$$

$k_{j\ell}$ ← coeficienți

Taylor $y(x_{i+1}) = y(x_i) + h \cdot f(x_i, y(x_i)) + \frac{h^2}{2!} y''(x_i) + \dots$

RK_1 = Euler $\quad k_{10} = 1$

RK_2

$y_{i0} = y_i$

$y_{i1} = y_i + h \cdot k_{10} f(x_{i0}, y_{i0})$

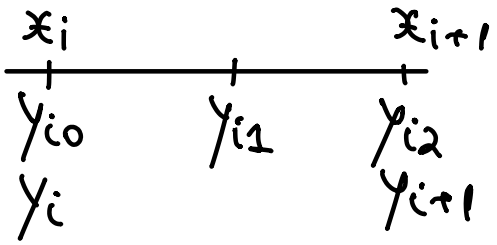
$y_{i2} = y_i + h k_{20} f(x_{i0}, y_{i0}) + h k_{21} f(x_{i1}, y_{i1})$

$x_{i1} = x_i + u_1 h$

$k_{10} = u_1$

$k_{20} = 1 - \frac{1}{2u_1} \qquad k_{21} = \frac{1}{2u_1}$

$y_{i2} \longrightarrow y_{i+1}$



Notons $\alpha_1 = \alpha$

$$y_{i+1} = y_i + \left(1 - \frac{1}{2\alpha}\right) k_1 + \frac{1}{2\alpha} k_2 \; ; \; y_0 = \lambda$$

$$k_1 = h \cdot f(x_i, y_i)$$

$$k_2 = h \cdot f(x_i + \alpha h, y_i + \alpha k_1)$$

ex $\alpha = 1 \; ; \; \alpha = 1/2 \; ; \; \alpha = 2/3$

$$\begin{cases} y_{i+1} = y_i + \dfrac{k_1 + k_2}{2} \\ k_1 = h f(x_i, y_i) \\ k_2 = h f(x_i + h, y_i + k_1) \end{cases}$$

RK4

$$\begin{cases} y_{i+1} = y_i + \dfrac{k_1 + 2k_2 + 2k_3 + k_4}{6} \\ k_1 = h f(x_i, y_i) \\ k_2 = h f\left(x_i + \dfrac{h}{2}, y_i + \dfrac{k_1}{2}\right) \\ k_3 = h f\left(x_i + \dfrac{h}{2}, y_i + \dfrac{k_2}{2}\right) \\ k_4 = h f(x_i + h, y_i + k_3) \end{cases}$$

RK_{gp} → grad de valabilitate
↘ ordin

$ode_{23} = RK_{23}$ = valabil pt. polinoame grad. max. 3

(4) Metode multipas

$$y_{k+1} = \sum_{i=0}^{u} \alpha_i y_{k-i} + h \sum_{i=-1} \beta_i f_{k-i} \qquad f_k = f(x_k, y_k)$$

$i = -1 \quad f_{k+1} = f(x_{k+1}, y_{k+1})$

metoda explicită $\beta_{-1} = 0$

metoda implicită $\beta_{-1} \neq 0$ (M. predictor-corector)

Problemă

Determinați formula implicită pentru

$$y_{k+1} = y_k + h(\beta_0 f_k + \beta_1 f_{k-1} + \beta_2 f_{k-2})$$

$x_k = x_0 + kh$ ⇓ $x_k = k$ translație $x_0 = 0$ scalare $h = 1$ $f = y'$

$y(x) = 1 \qquad 1 = 1$

$y(x) = x \qquad k+1 = k + \beta_0 + \beta_1 + \beta_2 \Rightarrow \beta_0 + \beta_1 + \beta_2 = 1$

$y(x) = x^2 \qquad (k+1)^2 = k^2 + 2\beta_0 k + 2\beta_1 (k-1) + 2\beta_2 (k-2)$

$$\Rightarrow \beta_1 + 2\beta_2 = -\frac{1}{2}$$

$y(x) = x^3 \qquad (k+1)^3 = k^3 + 3\beta_0 k^2 + 3\beta_1 (k-1)^2 + \beta_2 (k-2)^2$

$$\Rightarrow \beta_1 + 4\beta_2 = \frac{1}{3}$$

$$\Rightarrow 2\beta_2 = \frac{1}{3} + \frac{1}{2} \Rightarrow 2\beta_2 = \frac{5}{6} \Rightarrow \beta_2 = \frac{5}{12}$$

$$\beta_1 = -\frac{1}{2} - 2\cdot\frac{5}{12} = -\frac{1}{2} - \frac{5}{6} = \frac{-3-5}{6} = \frac{-8}{6} = \frac{-4}{3}$$

$$\beta_0 = 1 - \frac{-4}{3} - \frac{5}{12} = \frac{12+16-5}{12} = \frac{23}{12}$$

$$y_{k+1} = y_k + \frac{h}{12}\left(23f_k - 16f_{k-1} + 5f_{k-2}\right)$$

Luăm $y = x^4 \qquad f = y'$

$$(n+1)^4 = u^4 + \frac{1}{12}\left(23\cdot 4u^3 - 64(u-1)^3 + 54(u-2)^2\right)$$

pt $u=0$ avem $1 = \frac{1}{12}(64+216)$ (F)

Problemă

Determinați formula implicită pentru

$$y_{k+1} = y_k + h(\beta_{-1}f_{k+1} + \beta_0 f_k)$$

$y(x) = 1 \qquad 1 = 1$

$y(x) = x \qquad (k+1) = k + \beta_{-1} + \beta_0 \Rightarrow \beta_{-1} + \beta_0 = 1$

$y(x) = x^2 \qquad (k+1)^2 = k^2 + 2\beta_{-1}(k+1) + 2\beta_0 k$

$$2\beta_{-1} = 1 \Rightarrow \beta_{-1} = \frac{1}{2} \Rightarrow \beta_0 = \frac{1}{2}$$

$$y_{k+1} = y_k + \frac{h}{2}\left[f_{k+1} + f_k\right]$$

$$f_{k+1} = f(x_{k+1}, y_{k+1})$$

(5) Metoda predictor - corector

metodă explicită

metodă implicită

$$\begin{cases} y_{k+1}^{(p)} = y_k + \frac{h}{12}(23f_k - 16f_{k-1} + 5f_{k-2}) \\ y_{k+1}^{(c)} = y_k + \frac{h}{2}\left(f(x_{k+1}, y_{k+1}^{(p)}) + f_k\right) \end{cases}$$

$$y_{k+1} = y_{k+1}^{(c)}$$

$$y'' = f(x, y, y') \qquad y'' = x + y' + y^2 + 2$$

notăm $\begin{cases} y' = u \\ u' = f(x, y, u) \end{cases}$

$$y(x_0) = \lambda$$
$$u(x_0) = y'(x_0) = dat$$

$$\begin{cases} y_{i+1} = y_i + h\,U_i \\ U_{i+1} = U_i + h f(x_i, y_i, U_i) \end{cases} \qquad \begin{matrix} y_0 = \lambda \\ U_0 = \delta \end{matrix}$$